Loan Approval Prediction using Machine Learning (Python Project)

Introduction

Loan Approval Prediction is a machine learning classification project used to predict whether a loan application will be approved or rejected based on applicant details. Machine learning helps banks automate the loan approval process by analyzing applicant information such as income, credit score, loan amount, and employment status.

Objective

• Predict loan approval status.
• Reduce manual work in banks.
• Improve decision making using machine learning.

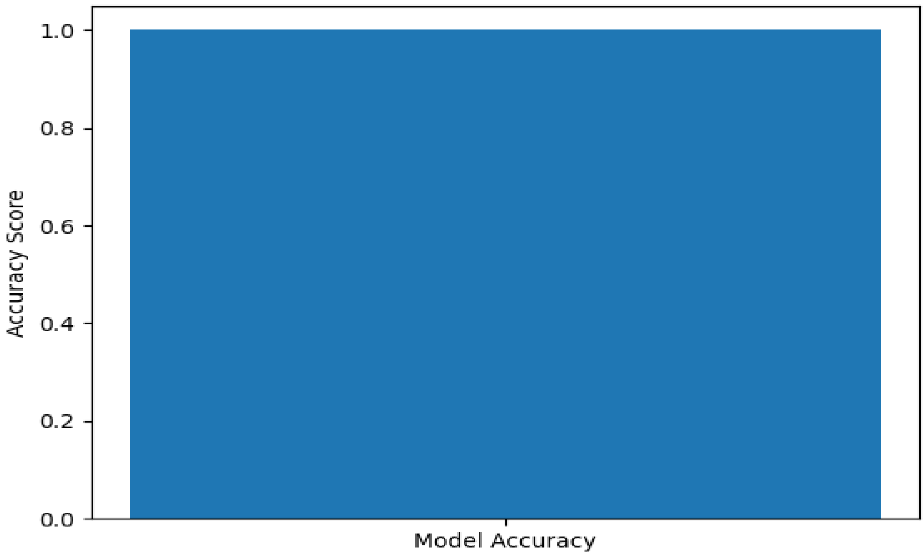
Algorithms Used

• Decision Tree
• Random Forest
• Logistic Regression

Python Implementation

```
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score

data = {
 'Income':[50000,60000,25000,80000,30000,70000],
 'CreditScore':[700,720,650,800,600,750],
 'LoanAmount':[20000,25000,12000,30000,15000,28000],
 'EmploymentStatus':[1,1,0,1,0,1],
 'LoanApproved':[1,1,0,1,0,1]
}

df = pd.DataFrame(data)

X = df[['Income','CreditScore','LoanAmount','EmploymentStatus']]
y = df['LoanApproved']

X_train,X_test,y_train,y_test = train_test_split(X,y,test_size=0.2)

model = DecisionTreeClassifier()
model.fit(X_train,y_train)

predictions = model.predict(X_test)

print("Accuracy:",accuracy_score(y_test,predictions))
```

Visualization



Sample Output

Model Accuracy: 1.0

Conclusion

The Loan Approval Prediction system demonstrates how machine learning can assist banks in making faster and more reliable loan approval decisions. By analyzing applicant data such as income, credit score, loan amount, and employment status, the model can predict whether a loan should be approved or rejected. This system can be further improved by training with larger real-world datasets.